

CLAVE 10F-04 e(d)

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$$v_{0x} = 15 \cos 37^\circ$$

$$v_{0y} = 15 \sin 37^\circ$$

H_{max}:

$$y_f = 15 \sin 37^\circ t - \frac{1}{2} g t^2$$

$$v_{fy} = 15 \sin 37^\circ - g t = 0$$

$$\frac{15 \sin 37^\circ}{g} = t \rightarrow t = 0,92 \text{ seg}$$

$$y_{\max} = 15 \sin 37^\circ (0,92 \text{ s}) - \frac{1}{2} g (0,92)^2$$

$$\boxed{y_{\max} = 4,88 \text{ m}}$$

X_{max}:

$$x_f = 15 \cos 37^\circ t$$

$$0 = 15 \sin 37^\circ t - \frac{1}{2} g t^2$$

$$0 = t (15 \sin 37^\circ - \frac{1}{2} g t) \rightarrow \frac{2 \cdot 15 \sin 37^\circ}{g} = t \rightarrow \underline{t = 1,84 \text{ s}}$$

$$x_{\max} = 15 \cos 37^\circ \cdot (1,84 \text{ s})$$

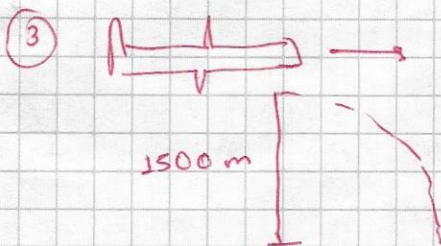
$$\boxed{x_{\max} = 22,07 \text{ m}}$$

$$\textcircled{2} \quad y(t) = v_{0y} t - \frac{1}{2} g t^2 = 25t - 4,9t^2 \rightarrow y(2) \Rightarrow \boxed{y(2) = 30,4 \text{ m}}$$

$$x(t) = v_{0x} t = 10t \rightarrow \boxed{x(2) = 20 \text{ m}}$$

$$v_x(t) = 10 \text{ m/s} \rightarrow \boxed{v_x(2) = 10 \text{ m/s}}$$

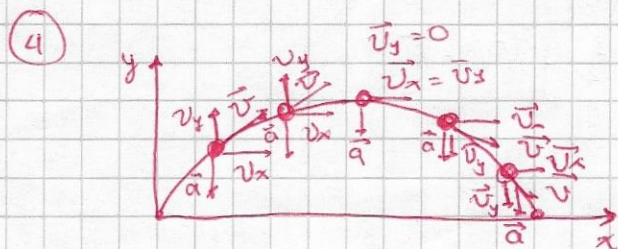
$$v_y(t) = v_{0y} - g t = 25 - 9,8t \rightarrow \boxed{v_y(2) = 5,4 \text{ m/s}}$$



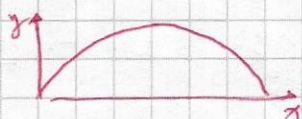
$$y_f = y_0 + v_{0y}t - \frac{1}{2}gt^2$$

$$1500 = 4,9 t^2$$

$$\sqrt{\frac{1500}{4,9}} = t \rightarrow t = 17,5 \text{ seg}$$



5) Parabólico:



$$y_0 = 0$$

$$y_{\max} \neq y_0$$

$$v_{0y} \neq 0$$

Semiparabólico:



$$y_0 \neq 0$$

$$y_{\max} = y_0$$

$$v_{0y} = 0$$

Ej: Patear un balón en un campo de fútbol.

Ej: Lanzar una piedra desde un edificio verticalmente.